

ESC101 20259 Design Report Independent Assessment Tool and Grading Rubric

Team 6

Assessor JP/LS

Grade 18

Submission requirements - Highlight those violated.

- 1. **Shall** be submitted as a single PDF file through Quercus.
- 2. **Shall** use a recognized reference format.
- 3. **Should** be in 11-pt serif font with 1.1-line spacing and 1.0-inch margins on standard letter-sized paper.
- 4. **Shall not** exceed 3000 words of text, including headings, tables, and captions, excluding any title-page content, table of contents, references, and appendices; and **shall** include a word count at the top of the first page.
- 5. **Shall** include relevant extracts from any used references in an appendix titled "Source Extracts."
- 6. **Shall** include the original design brief that provided the initial framing for the opportunity in an appendix.

Content requirements - Highlight those violated.

- 1. **Shall** be a standalone document, including appendices, that includes all of the information necessary for assessment and evaluation without the reader needing to locate other documents.
- 2. **Shall** recommend one (1) credible design concept that addresses the framed opportunity.
- 3. **Shall** document comparisons of the performance of the recommended design concept against at least three (≥3) other credible candidate design concepts.
- 4. **Shall** integrate a Pugh Chart.
- 5. **Shall** justify three (3) key design decisions in the recommended design, using a combination of verification (testing), prototypes, research, and/or calculation.
- 6. **Shall** include up to three (≤3) minutes of video content, linked in the relevant section of the design report, to represent a least one (≥1) key aspect of your design work (i.e., a key feature/working principle of (a) design concept(s), the verification of (a) design concept(s), etc.).
- 7. **Shall** document using at least one (≥1) prototype in verification activities.

Evaluation criteria	Unacceptable	Satisfactory	Good	Outstanding
Quality and credibility of engineering argument • Emphasis on justification of claims through credible use of engineering-appropriate sources	□ An over-reliance on opinion, description, or "it's obvious" statements. A lack of evidence throughout. Variety of evidence used is limited, with over-reliance on sources that are lacking in one or more of currency, relevance, authority, accuracy, or compatible purpose.	□ Claims appear credible at first glance, with most supported by evidence. A variety of resources were used (textbooks, handbooks, reviews, direct testing, interviews...). Some evidence may be inappropriate or weak for the claim being made.	✓ Claims are supported by credible evidence throughout. Interpretive claims are supported by more substantial evidence. Personal considerations are supported with attached evidence.	□ As per Good + resources are used credibly, with qualifiers and triangulation regularly applied. Acknowledgements/analysis of possible {weaknesses, risk, error, omissions} are provided.
Quality of the framing, including background, engineering requirements, and evaluation criteria • Background necessary to the understanding of the opportunity is included • Framing of the opportunity identifies the scope and important perspectives • The requirements and evaluation criteria clearly define the design space	□ Background and/or framing are missing or inadequate Requirements and/or Evaluation criteria missing critical components (e.g. aspect, metric) or present but do not allow alternative designs to be evaluated (e.g. metrics are unconnected to aspect), and/or requirements are not credible (e.g. constraints come from contrived numbers) Requirements/EC make little or no reference to relevant standards, handbooks, DfXs, or stakeholders.	□ Background is present, but descriptions may be in broad, general terms. Scope and important perspectives may be vague or generic. Requirements and Evaluation criteria allow most alternative designs to be evaluated and compared, but may not be robust or fine-tuned enough to evaluate a broad range of potential designs Requirements/EC founded in some credible basis (standards, guidelines, DfX definitions, handbooks, or stakeholders.)	✓ Background and framing clearly describe the opportunity and provide context for the requirements and evaluation criteria Requirements and evaluation criteria are well supported by evidence, including research beyond the stakeholder statements into the context of the opportunity, research that supports engineering considerations of chosen DfXs; metrics are informed by relevant codes and/or standards Codes and standards are specific to the opportunity, as applicable, rather than general. There is clear traceability back to stakeholders for most Req/EC.	✓ As per Good + requirements and evaluation criteria are internally consistent from aspects/property through metrics to constraints and criteria, such that a team can work with the whole set. Uses of codes, standards, DfX are not only appropriate but are modified to be made usable for Engineering Science students in their 1 st term of study who have limited time to complete their design activities. All requirements and EC have clear traceability back to stakeholder needs.
Quality of the application of engineering tools Tools may include tools for: exploring (diverging); assessing (converging); representing (e.g. prototyping); and verifying	□ Report shows a lack of evidence of tool use, using inappropriate tools, using tools inappropriately, or using tools without gaining insight from them.	□ Report documents standard use of a limited range of tools. Tools are handled appropriately.	✓ The tools were chosen with consideration given to the information they can provide, well used to gain insight and promote informed diverging and/or decision-making (converging).	□ As per Good + a diversity of tools/approaches are used in combination to triangulate across different approaches and enable the team to apply judgment.
Quality of the recommended concept, as demonstrated through verification against the requirements and assessment using the evaluation criteria	□ Recommended concept evokes immediate and warranted scepticism due to issues related to (e.g.) thermodynamics (perpetual motion), materials (unobtainium), un-usability, etc. Recommended concept is at a level of development expected at Alpha.	□ The recommended concept has been verified using one or more approaches; the tests may be flawed, calculations overly reduced, or the concept may still have problems that prevent it from meeting all requirements. Recommended concept appears more developed than something seen at Alpha.	□ The recommended concept meets the team's prioritized requirements through the development of one or more design decisions to more completely meet requirements. Verification demonstrates that the design decisions were appropriate for the concept. The recommended concept has been demonstrated to better meet the evaluation criteria than other concepts and/or reference designs	✓ As per Good + the design shows a unique approach to the opportunity. Design decisions are clearly in service of meeting requirements and improving the design along evaluation criteria. Design decision tradeoffs are acknowledged and prioritization of improvements reflect the stakeholders' and designers' expectations and values
Quality of the verification process of the recommended design concept as represented by appropriate prototypes that were used in testing	□ No, or minimal evidence of verification through (e.g.) prototype testing, calculations, research, etc. Verification data collected but use of or consideration of results demonstrates lack of thought and due diligence. Measurement processes do not address a metric linked back to the opportunity and its stakeholders. Strengths and limitations of the measurement processes and test protocols are not discussed.	□ Verification data generated from multiple approaches. Testing data collected but justification for protocol or use of data may be awkward. Measurement processes addresses a metric that may not be the most important, but can still be argued as relevant to the opportunity and its stakeholders. Basic strengths and limitations of verification data discussed, though discussion may be imbalanced or superficial.	✓ Verification data used appropriately to guide design, based on your (thorough) discussion of the strengths and limitations of the data, your measurement process, test protocol, and prototype(s). Both the measurement processes and the metric(s) that it address are well-justified with research, logic, or other strong evidence.	□ As per Good + Verification data is effectively generated to support critical metrics and decisions in the design concept. Where relevant, measurement process credibly augments 'standard' or more commonly-used processes with your own modifications based on your context and prototype.
Quality of the document as a design report • Allows reader to find information quickly • Uses structure to aid readability • Structure responds to content	□ Structure and tone used inappropriate to genre. Introduction lacks some elements. Arrangement of information (including sub-headings) within the report may be illogical or confusing. Paragraphs lack clear claims, support for claims, or organization.	□ Structure and tone appropriate to genre. Introduction contains all elements, although some may be generic. Arguments and headings follow a logical structure. Structural elements such as bullet lists or tables are present, but may be poorly used or over-used.	✓ Introduction is efficient, clear, and specific to establish purpose. Headings and sub-headings provide a clear structure for a reader. Structural elements are used appropriately and aid in clarity and understanding.	□ As per Good + the document guides readers through the report structurally and conceptually. Readers can understand the central claims of the document by skimming the sub-headings or topic sentences.
The coherence and clarity of written, visual, and video communication • Visual and text are chosen and integrated to clearly communicate meaning • Uses specific language of engineering and design • Video content is meaningful and offers an effective representation to suit its purpose	□ No visuals present or are all relegated to the appendix. Visuals lack captions. Or visuals are illegible or irrelevant and thus not useful to enhance the document's message. Language errors significantly detract from the readability and professionalism of the report. Video content missing, insufficient, excessive, ineffective for the format, or does not offer a meaningful representation for its purpose.	□ Visuals are present in the document (and captioned), but may not be integrated into text. Visuals may be lacking in areas where they would aid in a reader's understanding. Writing is clear, but may be repetitive or show a disruptive preference for flourish over concision. Video content is clear, of reasonable length and scope, and conveys at least one aspect of the design work, though it may not be ideal for its purpose or a different choice of video content may have been more beneficial for the design report overall.	✓ Visuals referred to and integrated with text. Choice of visuals improves understanding and clarity. Writing is clear, concise, and demonstrates awareness of specific engineering and design language. Coherence is maintained throughout paragraphs and arguments. Video content is an effective representation for its purpose in the design report and conveys one or more clear and complete ideas that are best communicated in this format.	□ As per Good + Visuals include appropriate guidance for the reader. Writing style and structure demonstrates nuanced control of language through appropriate use of qualifiers in arguments, clear and concise expression, and precision in word choices. Video content enhances the overall effectiveness of the design report by making optimal use of the format for communication.
Coherence of the vision of the team's work represented in the writing style of the document	□ Document sections are clearly authored by different writers with no attempt to establish an overall style or approach. Statements in one section may contradict statements in another section.	□ Different sections of the document may differ in style or use of terminology but do not contradict one another. Attempts have been made to link different sections with reference to relevant information in other sections although perspective may vary. Vocabulary is used consistently throughout the document.	✓ A recognizable voice is present, although inconsistently, throughout the document. Different sections reference one another to establish evidence for claims and to establish a line of logical reasoning throughout the entire document.	□ As per Good + the document flows smoothly from one section to the next without breaks in the line of logical reasoning, use of evidence, reference to actions the team took, or descriptions of product/process.

Holistic assessment		
Key things that contributed to the quality of your design report	Key things that detracted from the quality of your design report	Key things to think about for next time
- Great job in incorporating very descriptive headings for each section of the report. With a glance of the table of contents, the reader is able to quickly assimilate the key ideas to be presented in the design report. - Demonstrated an ability to synthesize information across different sections by referencing one another to establish support for claims in the text. This was most evident when utilizing pairwise comparison and Pugh charts to inform a recommended design for the opportunity. - Great research + justification for the requirements + ECs present in the report - good job using multiple pugh charts + discussing tradeoffs between the solutions	- There may be instances of disconnect between what is explained in the text and what is shown as evidence. Please be sure to proofread and ensure the evidence aligns with what is being explained so as to not diminish the quality of the report. - It was not entirely clear how the ball would 'push ink into the whiteboard'. The whiteboard should be designed to be non-porous so ink should not theoretically be pushed into it? Perhaps some elaboration and further support on this idea is necessary. This can connect back to the notes about the board mentioned in the 'working environment' section. - Consider highlighting via annotations or something similar on what to pay attention to in the video in addition to the labelling of what testing it depicts. Should the viewer be paying attention to the angle of the wrist? - don't forget to include limitations to the protocols listed	- Though the video was somewhat helpful to understand the prototypes in action, in the baseline test of the subject, no smudging occurred -- if this is the case, then why is this a splartz? - a measurement matrix may be helpful to summarize key results (from testing + research) to better support the pugh chart